

Questions of 2019

Answer to the Q.No: 5 (a)

Inverse Relation: An inverse relation of a relation is a set of ordered pairs which are obtained by interchanging the first and second element of the ordered pairs of the given relation.

If, $R = \{(x, y) : x \in A \text{ and } y \in B\}$ then $R^{-1} = \{(y, x) : y \in B \text{ and } x \in A\}$

Answer to the Q.No: 5 (b)

(i) Given that,

$$R = \{(x, y) : x + 3y = 12\}$$

x	9	6	3	0	12
y	1	2	3	4	0
x, y	9, 1	6, 2	3, 3	0, 4	12, 0

$$\therefore R = \{(9, 1), (6, 2), (3, 3), (0, 4), (12, 0)\}$$

$$(ii) R = \{(9, 1), (6, 2), (3, 3), (0, 4), (12, 0)\}$$

$$\therefore \text{Domain of } R = \{9, 6, 3, 0, 12\}$$

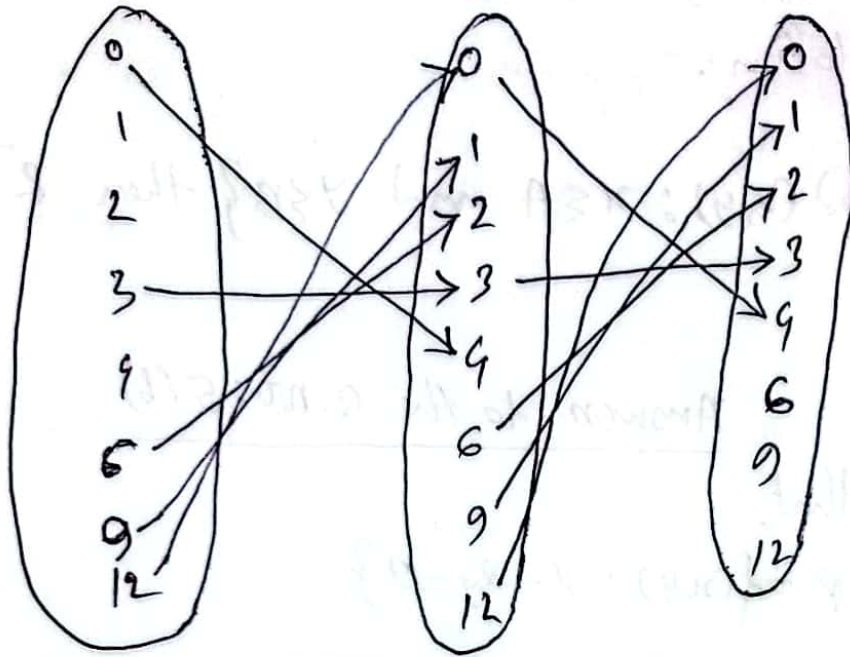
$$\text{Range of } R = \{1, 2, 3, 4, 0\}$$

$$\text{Inverse of } R \text{ of } R^{-1} = \{(1, 9), (2, 6), (3, 3), (4, 0), (0, 12)\}$$

(iii) Here

$$R = \{(9,1), (6,2), (3,3), (0,4), (12,0)\}$$

$$\therefore R \circ R =$$



$$\therefore R \circ R = \{(3,3), (12,6)\}$$

Answer to the Q. NO: 5(c)

Given that, $A = \{a, b, c\}$ and

$$R = \{(a,a), (a,b), (b,c), (c,c)\}$$

Reflexive: R is not reflexive relation since $(b,b) \notin R$

Symmetric: R is not ~~reflexive~~ symmetric relation since $(a,b) \in R$ But $(b,a) \notin R$ and $(b,c) \in R$ But $(c,b) \notin R$.

Transitive: R is not transitive relation since $(a,b) \in R$ and $(b,c) \in R$ But $(a,c) \notin R$.

Answer to the Q. no: 6(a)

Hasse Diagram: A hasse diagram is a graphical representation of the relation of elements of a partially ordered set upward orientation.

Given poset = $(\{1, 2, 3, 4, 6, 8, 12\}, \preceq)$

$$\therefore R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 6), (1, 8), (1, 12), (2, 2), (2, 4), (2, 6), (2, 8), (2, 12), (3, 3), (3, 6), (3, 12), (4, 4), (4, 8), (4, 12), (6, 6), (6, 12), (8, 8), (12, 12)\}$$

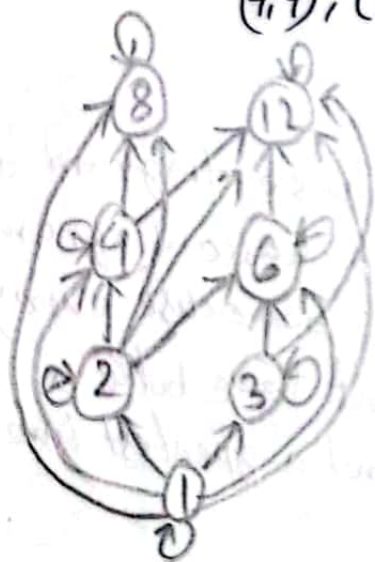


Fig (A)



Fig (B)

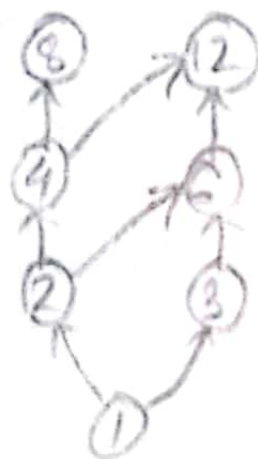


Fig (C)

Step-1: The directed graph of the relation is shown in Fig (A)

Step-2: Remove all self-loop as shown fig(b).

Step-3: Delete all the edges implied by transitive property. Thus are $(1, 4), (1, 6), (1, 8), (1, 12), (2, 8), (2, 12), (3, 12)$.

Arranged all edges to print upward the hasse diagram. The Resulting hasse diagram is shown in fig (c).

Answer to the Q. no: 6 (b)

Lattice: A Lattice is a poset, a partially ordered in which every pair of elements has both a least upper bound and a greatest lower bound.

Given poset: $(\{1, 2, 3, 4, 6, 8, 12\}, |)$

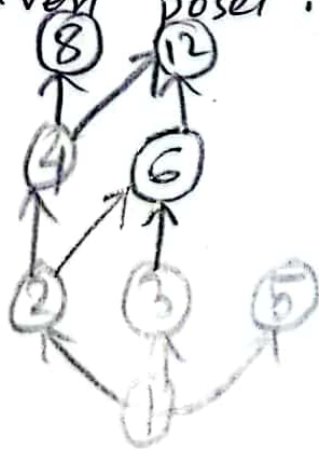


Fig - (a)

Here, fig (a) is hasse diagram of given poset. By this hasse diagram it's not a lattice because every pair of elements doesn't have both a least upper bound and a greatest lower bound.

Again,

Given poset: $(\{1, 2, 4, 8, 16\}, |)$



Here, fig (c) is hasse diagram of given poset. By this hasse diagram it's a lattice cause here every pair of elements have both a least upper bound and a greatest lower bound.

Answer to the Q. no: 6 (c)

Given poset: $(\{3, 5, 9, 15, 24, 45\}, |)$
 $\therefore R = \{(3, 3), (3, 9), (3, 15), (3, 24), (3, 45), (5, 5), (5, 15), (5, 45), (9, 9), (9, 15), (9, 45), (15, 15), (15, 45), (24, 24), (45, 45)\}$

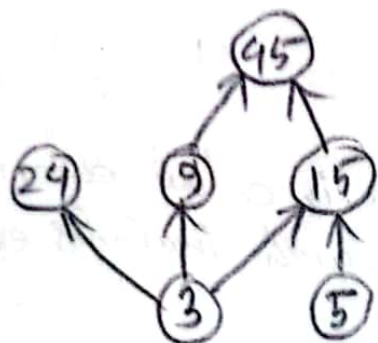


Fig (a)

Here, fig (a) is hasse diagram of given poset.

Here,

(i) Maximal element = $\{24, 45\}$

(ii) Minimal element = $3, 5$

(iii) least upper bound of $\{3, 5\} = 15$

(iv) Greatest lower bound of $\{15, 45\} = 15$

Answer to the Q. no: 7 (a)

Bipartite Graph: Bipartite Graph is a set of graph vertices decomposed into two disjoint sets such that no two graph vertices within the same set are adjacent.

complete Bipartite Graph: It is a bipartite graph such that every pair of graph vertices in the two sets are adjacent.

Example:

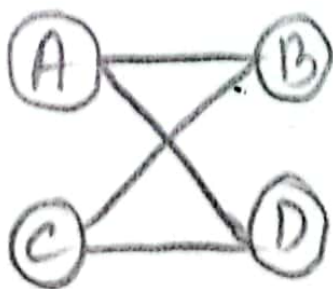


Fig (a): Bipartite graph

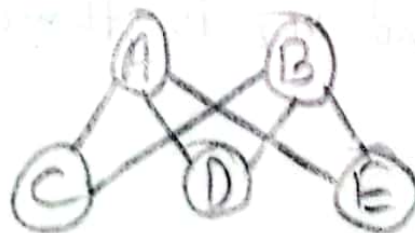


Fig (b): complete Bipartite

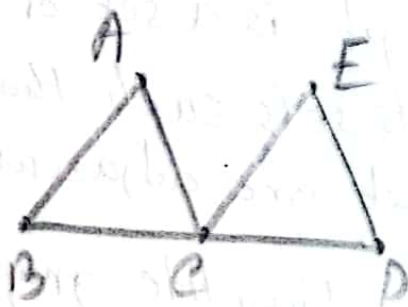
Answer to the Q. No: 7(b)

The given pair of graph is not isomorphic.
Because, here both graph has equal vertices, Nodes, Degree and mapping.

Answer to the Q. No: 7(c)

Eulerian Graph: A connected graph G is called eulerian graph if if there is a closed trail which includes every edge of the graph G .

Hamiltonian Graph: A connected graph G is called hamiltonian graph if there is a cycle which includes every vertex of G and the cycle is called hamilton cycle.



G_1



G_2

Here, G_1 is Eulerian Graph but not Hamilton Graph
And G_2 is Hamilton Graph but not Eulerian Graph.

Answer to the Q. no: 7(d)

let us consider,

$$K_n = 3$$

$\therefore$ Graph $K_{n,n}$ on $K_{3,3}$ is given below:

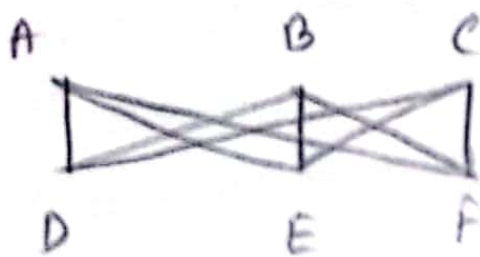


Fig - $K_{3,3}$

Here, sum of degree A and D is 6 which is equal to total number of nodes, thus it is a hamilton graph. so, it's proved that complete bipartite graph $K_{n,n}$ has hamilton cycle.

Questions of 2020

Answer to the Q. NO: 5 (a)

Binary Relation: A binary relation R on a single set A is a subset of $A \times A$. For two distinct sets, A and B , having cardinality of m and n respectively, the maximum cardinality of a relation R from A to B is mn .

An example of Binary Relation is the divides relation over the set of prime numbers and the set of integers.

The properties of relations: The properties of relations are

1. Reflexive ($a \sim a$)
2. Symmetric ($x \sim 2$ implies $2 \sim x$)
3. Transitive ($2 < 3$ implies and $3 < 5$ implies $2 < 5$)
4. Irreflexive ($2 < 3$ implies $2 \neq 3$)
5. Antisymmetric ($(a, b) \in R$ implies $(b, a) \notin R$).

Answer to the Q. NO: 5 (b)

Given that,

$$A = \{1, 2, 3, 4\}$$

(a, b) is in R if and only if $a > b$. So the relation

$$R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

Arrow diagram of R is

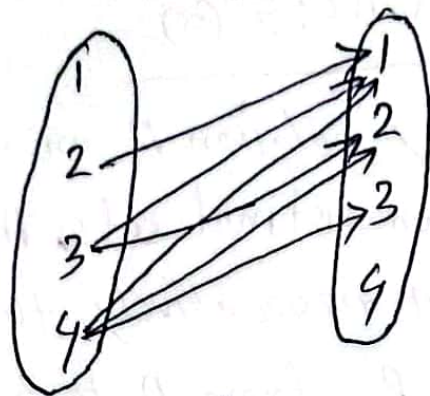


Fig: Arrow diagram of R .

Zero - One Matrix of R is

	1	2	3	4
1	0	0	0	0
2	1	0	0	0
3	1	1	0	0
4	1	1	1	0

Fig: Zero - One Matrix of R .

Diagram of R is,

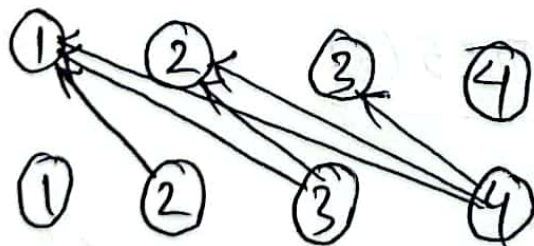


Fig: Diagram of R .

Answer to the Q.No: 5(c)

The number of reflexive relation on a set with the 'n' number of elements is given by $N = 2^{n(n-1)}$, where N is the number of reflexive relation and n is the number of elements in the set.

Example: An example of a reflexive relation is the relation "is equal to" on the set of real numbers, since every real number is equal to itself.

Answer to the Q.No. 6(a)

Lattice: A partially ordered set in which every pair of elements has both a least upper bound and greatest lower bound is called lattice.

Totally Ordered set: A poset POSET in which every element are comparable is called totally ordered set or TOSET or a chain.

Ex → ① $(\mathbb{R}, \text{usual } \leq)$ is a chain or TOSET. As, $2 \leq 3$

② $(\mathbb{N}, \text{divisibility})$ is not a TOSET.

as, 2×3 & 3×2

③ $(P(X), \subseteq)$ is not a TOSET.

$X \neq \emptyset$

As, $P(X) =$
 $X = \{a, b\}$

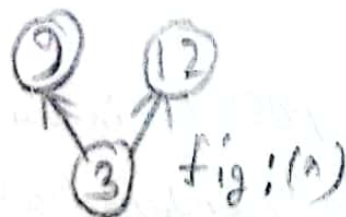
$$\therefore P(n) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

Here, $\{a\} \not\subseteq \{b\}$ and $\{b\} \not\subseteq \{a\}$

So, By definition of ~~POSET~~ lattice and TOSET we can say every TOSET is lattice but every lattice is not TOSET. Because, for making TOSET we need to comparable set which already in lattice. But so, Every totally ordered set is lattice.

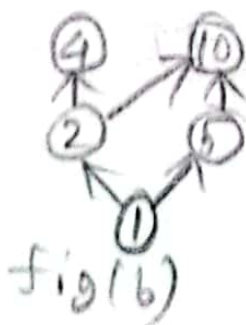
Answer to the Q. NO: 6(b)

Given poset (1) : $\{3, 9, 12\}$



Hasse diagram of given poset fig (1).

Given poset (2) : $\{1, 2, 4, 5, 10\}$



Here, fig (2) is hasse diagram of poset

So, To find greatest lower bound and least upper bound we need a subset but here no subset is indicate. so, we can't find the greatest lower bound

Answer to the a.m. 6(c)

Given,

$$A = \{1, 2, 3, 4\} \text{ and}$$

$$R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

$$M_R = W_0 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step-1:

$$\begin{array}{l} 1^{\text{st}} \text{ col} \rightarrow \emptyset \\ 1^{\text{st}} \text{ row} \rightarrow \{2, 3, 4\} \end{array} \therefore R = \{\emptyset\}$$

$$W_0 = W_1 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step-2: 2nd column $\rightarrow \{1\}$

2nd row $\rightarrow \{3, 4\}$

$$\therefore R = \{(1, 3), (1, 4)\}$$

$$W_0 = W_1 = W_2 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step-3: 3rd column $\rightarrow \{1, 2\}$

3rd Row $\rightarrow \{4\}$

$$\therefore R = \{(1, 4), (2, 4)\}$$

$$W_3 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step-4: 4th column $\rightarrow \{1, 2, 3\}$

4th Row $\rightarrow \{\emptyset\}$

$$\therefore R = \{\emptyset\}$$

$$\therefore W_3 = W_4 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore R^+ = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$ is a transitive closure of the Relation.

Answer to the Q.No: 7 (a)

Pseudograph: A graph that has either self-loops or multiple edges or both, is called pseudo graph.

Multigraph: A multigraph is a graph which is permitted to have multiple edges is called multigraph. But no self-loops.

Difference between pseudograph and multigraph:

A multigraph is a graph which is permitted to any coding algorithm have multiple edges that is edges that any coding algorithm have the same end nodes. Thus two any coding algorithm vertices may be connected by more than any coding algorithm one edges and.

A pseudograph is a multigraph that is any coding algorithm permitted to have loops, so the differentiating factor is that a any coding algorithm self-edges, while pseudograph can have any coding algorithm both.

Answer to the Q. No: 7(b)

Adjacency matrix of these given graphs are given below:

(i)

	A	B	C	D
A	1	1	0	1
B	0	0	0	1
C	0	0	0	0
D	1	0	1	0

(ii)

	A	B	C	D	E	F
A	0	1	0	0	0	0
B	1	0	1	1	0	0
C	0	0	1	1	0	0
D	0	1	1	1	0	1
E	0	0	0	0	0	1
F	0	0	0	1	1	0

Answer to the Q. no: 7(c)

Incidence Matrix: Incidence matrix is that matrix which represents the graph such that with the help of that matrix we can draw a graph.

Example:

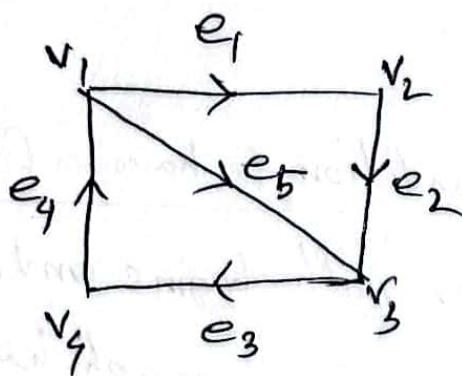


fig: (a)

$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Incidence Matrix of fig(a).

And,

Incidence Matrix of Given graph is:

$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Answer to the Q. NO: 8 (a) A

Euler path: An euler path is a path that uses every edge of graph exactly once.

one example of euler circuit for this graph is A, E, A, B, C, B, E, C, D, E, F, D, F, A. This is a circuit that travels over every edges once and only once and starts and ends in the same place.

Necessary and sufficient condition to have a Euler path are

An euler circuit is an euler path begins and ends at the same vertex. Euler theorem: If a graph has more than 2 vertices of odd degree then it has no Euler paths.

Answer to the Q. NO: 8 (b)

M-ary tree: The tree is called a full m-ary tree if every internal vertex has exactly m children. An m-ary tree with $m=2$ is called binary tree.

A rooted m-ary tree of height h is balanced, if all leaves are at level h or $h-1$.

Answer to the a.no: 8(c)

Spanning Tree: A spanning tree is a subset to a connected graph G , where all the edges are connected.

Example:

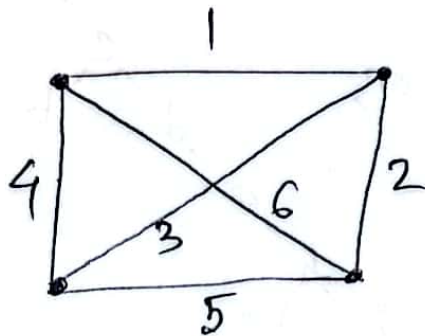


Fig: Spanning Tree.

and,

use DFS to find a spanning tree for the following graph
~~also~~ choosing the root a . The steps are:

From root or Node a to other nodes paths we have to check first. then, select any node from that path. Then, By these way have to go till last node which ~~can be~~ is connected. Then, we have to return back to the previous node ~~on check~~ and again check which node are not traversed. If any node is ~~find~~ found which is not traversed then go to that node and again follow the first rule. then when get the last not again return and check which node is not traversed. By these way we can complete our DFS.

DFS of spanning tree from root a.

